TUAN DO

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Hanoi, Vietnam

Summary

Applied Mathematics graduate with a strong background in stochastic processes and numerical methods. Awarded the prestigious PGSM Scholarship (FSMP) for graduate studies at the University of Paris Dauphine - PSL. Proven track record in bridging the gap between rigorous mathematical theory and practical applications in Finance and AI. Expert in developing probabilistic models and machine learning systems.

Education

- **University of Paris Dauphine – PSL [ ]** — *Upcoming Sep 2026*
  Master 1 Mathematics and Applications, major in Advanced Mathematics — Paris, France
  - **PGSM Scholarship**: Prestigious merit-based scholarship granted by The Fondation Sciences Mathématiques de Paris(FSMP).
- **WorldQuant University [ ]** — *April 2025 - Now*
  MSc in Financial Engineering — Online Program
- **Budapest University of Technology and Economics [ ]** — *Sep 2022 - Jan 2025*
  BSc in Mathematics, Specialization in Stochastic Processes — Budapest, Hungary
  - GPA: 4.68/5.0. **Valedictorian** of the faculty and the **first in history** to complete the program a semester earlier than expected.
  - Thesis grade: 5.0/5.0. Special Award at BME Student Scientific Conference (TDK).
  - Focus: Stochastic Processes, Probability and Statistics, Graph Theory, Deep Learning, Machine Learning, Numerical Methods, Mathematical Modeling, Theory of Algorithms, Data Science.

Research Experience

- **Vietnam Institute of Advanced Study in Mathematics[ ]** — *Aug 2025 - Nov 2025*
  Research Fellowship — Hanoi, Vietnam
  - Supervisor: Prof. Hoang Manh Tuan.
  - Research Thesis: A Modified SIS Epidemic Model with Application to Health Insurance Pricing
  - Investigated a modified version of the classical SIS model, performed dynamical analysis and numerical simulation to investigate the existence and asymptotic stability of the disease-free equilibrium points.
- **Institute of Mathematics - Vietnam Academy of Science and Technology [ ]** — *April 2025 - July 2025*
  Summer Research Fellowship — Hanoi, Vietnam
  - Supervisor: Prof. Can Van Hao.
  - Research Thesis: Mixing times of Markov chains. Derived bounds on total variation distance and analyzed the cutoff phenomenon using coupling and spectral gap techniques.
- **Budapest University of Technology and Economics [ ]** — *Dec 2024*
  Bachelor Thesis — Budapest, Hungary
  - Supervisor: Prof. Roland Molontay and Marcell Nagy.
  - Research Thesis: The Impact of Social Media Sentiment on Cryptocurrency Markets.
  - Skills: Sentiment Analysis, Recurrent Neural Network, Time series analysis, Natural Language Processing.

Work Experience

- **FiinGroup Joint Stock Company [ ]** — *May 2025 - Now*
  Middle Credit Risk Modeling — Hanoi, Vietnam
  - Developed and validated credit risk models for Forward-Looking PD term structure, including Survival Analysis (CoxPH, Gradient Boosting Survival, Kaplan-Meier) and Markov Chain approach with IFRS 9 standards.
  - Developed an AI-Driven Business Intelligence Search & Recommendation System combining SQL Agent and Hybrid Search(semantic search + keyword), enabling the system to perform automated schema inspection to identify and recommend comparable organizational entities with 24% higher accuracy.
  - Developed a crawling financial news system that autonomously crawls news from various sites and extracts useful information about organizations
- **BKK - Budapesti Közlekedési Központ [ ]** — *Sep 2023 - Jan 2024*
  Data Scientist Intern — Budapest, Hungary
  - Implemented various machine learning and deep learning models from sklearn and keras in predicting bike sharing demand. Leveraged insights to curate bike placing strategies for boosting company growth.
  - Developed a dashboard using GeoPandas, OSMnx, NetworkX, and Ipyleaflet to visualize the network of streets, accidents, and traffic congestion in Budapest.

PROJECTS

- **Cryptocurrency Prediction Based on Social Media** *Nov 2024*
  Pandas, Selenium, Tensorflow, statsmodels, Plotly, LLM []
  - Data Engineering: Scraped and processed a dataset of 40,000+ cryptocurrency news articles from Telegram channels using BeautifulSoup, Selenium, and TelegramAPI.
  - Sentiment Analysis: implemented Transformer-based models (BERT, RoBERTa, DistilBERT) to classify market sentiment and visualize correlations between public opinion and Bitcoin price movements.
  - Predictive Modeling: Developed and evaluated time-series forecasting models (ARIMA, Prophet, LSTM, GRU). Achieved a 0.09 MAPE (Mean Absolute Percentage Error) using LSTM and GRU architectures enhanced by sentiment features.

SKILLS

- **Programming:** Python, R, C++, MATLAB, SQL, LaTeX
- **Machine Learning & AI:** TensorFlow, Keras, Scikit-learn, LangChain, LLMs, Statsmodel, Optuna.
- **Data Engineering & Tools:** MySQL, SQLAlchemy, Git, GCP, Docker, Selenium, BeautifulSoup, Streamlit.
- **Language:** Vietnamese(Native), English(Professional Working Proficiency), French (A2), Hungarian (A2).

HONORS AND SCHOLARSHIPS

- **PGSM Scholarship**, *The Fondation Sciences Mathématiques de Paris(FSMP)* *April 2026*
- **Erasmus Mundus Joint Master of MSc in Interdisciplinary Mathematics Scholarship**, *European Union* *April 2026*
  - Prestigious merit-based scholarship for a fully funded 2-year MSc. Ranked 6/2465 applicants worldwide.
- **Erasmus Mundus Joint Master of MATHS-DISC Scholarship**, *European Union* *April 2026*
  - Top 15 students selected from over 1,600 applicants worldwide for a fully funded master program
- **Finalist at HackCX Hackathon**, *Techcombank and Hanoi University of Science and Technology* *June 2025*
- **Special Award at BME Student Scientific Conference (TDK)**, *Department of Finance, BME* *Nov 2024*
- **Certificate of Merit for Outstanding Academic**, *Embassy of the Socialist Republic of Vietnam in Hungary* *Feb 2024*
  - Awarded to students with excellent academic performance and great contribution to the Vietnamese Student Association in Hungary
- **Stipendium Hungaricum Excellence Award 2023 & 2024**, *Tempus Public Foundation, Hungary*
  - Awarded to top 3 students from each institution for outstanding academic performance and their contribution to the development of the international Stipendium Hungaricum community.
- **Honorable Mention Prize at BME Annual Mathematics Olympiad** *May 2024*
  - Awarded by the Faculty of Natural Science - BME and Morgan Stanley
- **Budapest Scholarship Research Program**, *Municipality of Budapest, Hungary* *Sep 2023*
  - Fully funded research program awarded to top 20 students with excellent background in research.
- **First Prize**, *Techainer AI Hackathon* *Oct 2022*
- **Stipendium Hungaricum Scholarship**, *Tempus Public Foundation - Hungary* *July 2022*
  - Prestigious merit-based scholarship covering full tuition fee and living expenses.

CONFERENCE AND SUMMER SCHOOL

- **School on Dimension Theory of Fractals**, *Alfréd Rényi Institute of Mathematics, Budapest, Hungary* *Aug 2024*
- **School: Optimal Transport on Quantum Structures**, *Alfréd Rényi Institute of Mathematics, Budapest, Hungary* *Sep 2022*

EXTRACURRICULAR ACTIVITIES

- **Scientific Researcher** *May 2022 - Jan 2024*
  Mathematics Modeling Vietnam, Hanoi, Vietnam
  - Participated in Mathematical Contest In Modeling (MCM).
  - Mentored and held training sessions for the Mathematics Modeling Vietnam Contest's competitors.
- **Academic Mentor** *April 2024*
  Vietnamese Students Association In Hungary, Budapest, Hungary
  - Organized consultations to answer questions regarding the scholarship's application.
  - Facilitated the academic and cultural integration of freshmen into the Hungarian education system.
- **Sports**
  - Champion of the Carritas League 2025 organized by former students at Chu Van An High School For The Gifted.
  - Third Place at The Vietnamese Society in Hungary Football Cup 2024.
  - Runner Up at The Vietnamese Society in Hungary 3x3 Basketball League 2024.

RESEARCH PUBLICATIONS

[1] Tuan Chau Do, Tien Thinh Le, Nguyen Trong Hieu, Manh Tuan Hoang. **A Modified SIS Epidemic Model with Application to Health Insurance Pricing, 2026**. arXiv:2601.02168 [math.DS]. Manuscript submitted for publication in *São Paulo Journal of Mathematical Sciences*.